



FOUR-COURSE SYSTEM

COURSE 3 - DATA ENGINEERING 2026

C3-054A SolarFleet Protected Review

Open only after the matching Gate attempt/evidence is frozen.

Artifact	C3-054A-REVIEW
Status	2026 - LEARNER EDITION
Phase	C3-04 Storage, File Formats & Lakehouse 13 lessons 34-46 focused hours
Prerequisite	C3-03 Gate PASS; Bridge warehousing/lakehouse mental models assumed, implementation depth taught here.
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House teaching rule: mental picture -> tiny example -> guided real file/table workflow -> break/fix -> independent changed task -> validation -> production reasoning -> evidence. A file extension or successful write is never sufficient proof of correct table state.



Expected current-state controls

After applying the supplied correction by latest version semantics, current-state row count is 6 and total reading_value across current rows is 245.20. Key reading_id remains unique. The added field exists only where supplied unless your explicit contract defines a default.

Required lakehouse properties

- Physical file/layout choice is justified by workload and does not claim a universal winner.
- Committed Delta version/log evidence is real.
- Schema change is classified and explicitly allowed/blocked.
- MERGE predicate matches the declared business grain and source uniqueness is proven.
- Historical version read names exact differences and retention dependency.
- Compaction is evidence-triggered; post-rewrite parity is explicit.
- Vacuum/retention preserves the declared recovery window.
- Delta/Iceberg recommendation states assumptions and reversal conditions.

Routing

PASS only when every required competency passes and zero critical failures remain. PARTIAL/FAIL -> targeted repair + changed micro-retest. A fresh full Gate B/C opens only by explicit assignment; failed attempts remain in history.